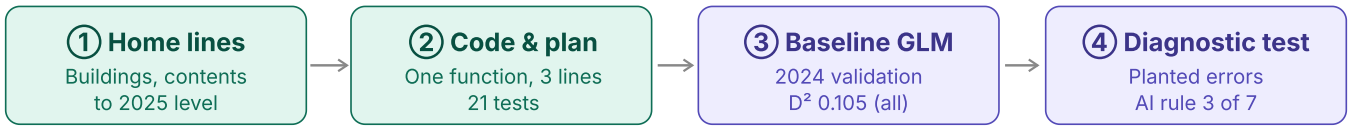
# Work Summary: October 18

IAA Hackathon · Sunday, October 18, 2026

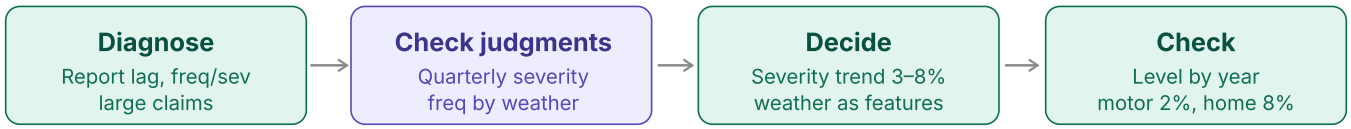
In one line: finished the 2025-level target for all three lines, built the first model, and tested a diagnostic rule proposed by AI.

## Overall flow



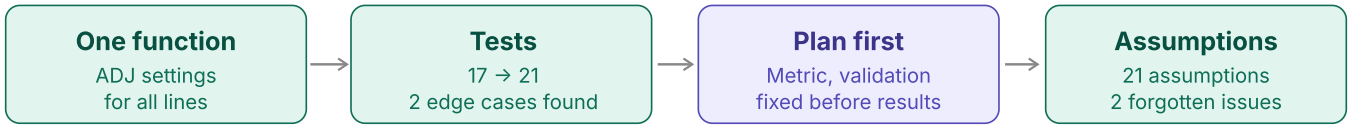

Green = done · Purple = key finding · Grey = next

## ① Buildings and contents adjustments



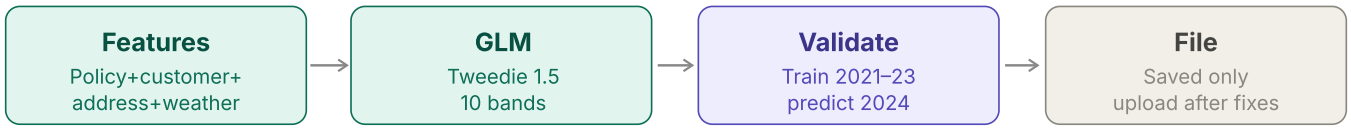

- Severity rises gradually (inflation), not as a step: quarterly medians climb steadily.
- Weather explains frequency strongly: subsidence 3.1x from driest to wettest quintile; rain, wind, wildfire 1.8–2.5x. 2025 weather exists for 100% of test addresses.
- Storm and fire year-to-year changes are not explained by weather (open). Fire and contents theft frequency trends to be decided by validation.
- Buildings cap 300,000 (same 99.9th percentile rule as motor); no cap for contents.
- Expected 2025 level incl. loading: motor ~564, buildings ~467, contents ~106 per exposure.
- Buildings 2024 is 7% high, mostly from the subsidence immaturity estimate (69% complete): the main buildings uncertainty.

## ② Code and modelling plan



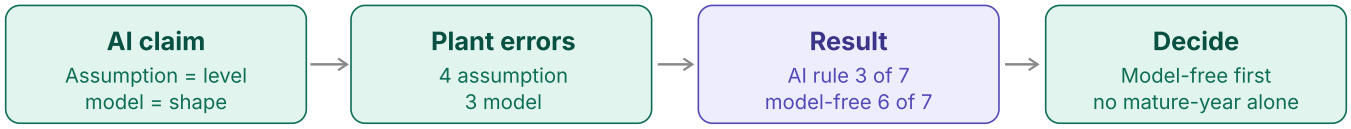

- Adjustments moved to per-line settings, so Week 2's new market only needs new settings.
- Modelling plan fixed before any result: Tweedie deviance (1.5), out-of-time validation, policy-grouped K-fold, no model choice by leaderboard.
- The assumption list resurfaced open-claim recoveries and size-dependent BI reform. Recoveries checked: 0.34% of motor, recorded, not adjusted.

## ③ Baseline GLM and first validation



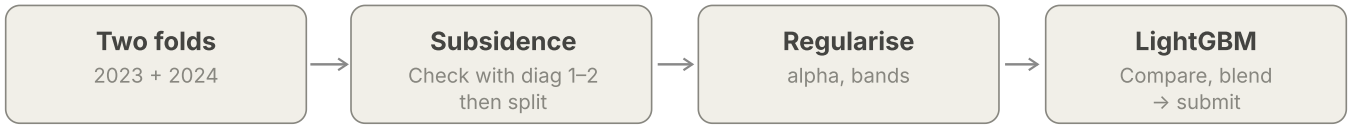

- $D^2$: motor 0.052, buildings 0.058, contents 0.062, all 0.105. Top decile costs 6–9x the bottom: risk is being separated.
- Motor is good (A/E 1.025, monotone lift). Buildings over-predicts by 7%, concentrated in high-risk homes: one-completeness-per-line assumption suspected.
- Contents and low deciles: relativities too spread (over-fitting), needs regularisation. Region A/E swings by year (regional weather events).
- 12 model assumptions judged: 3 violated, 5 partly, 4 untested. Submission postponed until fixes.

## ④ Testing AI's diagnostic rule (planted errors)




- Question chain: 'how do we tell assumption errors from model errors?' → 'how do we verify that claim itself?' → experiment.
- Why the rule failed (1): AI's level metric measured distance from 1, so errors offsetting the baseline bias looked like improvements (AI design error). Measured against the baseline, all four assumption errors are attributed correctly.
- Why (2): model errors also shift the level (dropping age moved 2024 A/E by 2.7%). The over-fitting case was too weak with 980k rows.
- Most useful outcome: the mature-year comparison mistook two model errors for immaturity, so tomorrow's buildings call must not rely on it alone.
- A rule noticed after seeing the results (assumption errors move A/E, model errors lower $D^2$) is kept as a hypothesis only.

## Next




Records: AI log entries 20–31 (8 questions from the team, 3 AI errors), Decision Log, data preparation summary PDF, GLM assumptions PDF, materials for Krista.